

German Electricity Load Forecasting: Time Series Modelling Case Study

1. Data preparation and exploratory analysis

The German load column DE_load_actual_entsoe_transparency was extracted from the 60-minute single-index OPSD file. After filtering to observations from 2015 onward, the filtered dataset retained 50,400 hourly values. These were resampled to daily and weekly mean loads. Weekly aggregation was used for most statistical models because it reduces high-frequency noise while preserving the annual demand cycle relevant for a two-year forecasting horizon.

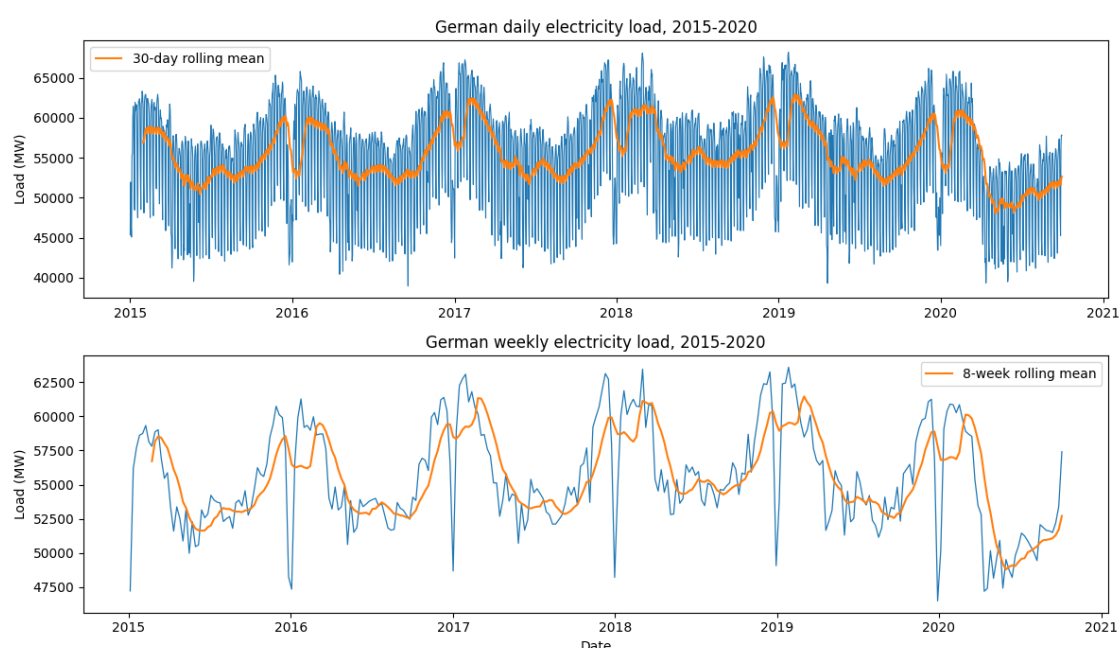


Figure 1. Daily and weekly German electricity load with rolling means, 2015-2020.

The exploratory plots show a strong repeating annual pattern, with higher winter loads and lower summer loads. The weekly series also shows structural disruption in 2020, consistent with an abnormal demand period. The daily decomposition confirmed a weekly intra-week component, while the weekly decomposition showed an annual seasonal component with period 52. This seasonality explains why the seasonal naive model is a difficult benchmark: simply repeating the last observed annual pattern already captures much of the load structure.

Series	ADF p-value	ADF stationary?	KPSS value	p-value	KPSS stationary?	Interpretation
Weekly level	0.0012	Yes	0.10	Yes	Yes	Level series is sufficiently stationary by both tests, but seasonal ACF remains.
First difference	<0.001	Yes	0.10	Yes	Yes	Stationary, but differencing may remove useful level information.
Seasonal diff. 52	0.00045	Yes	0.01	No	No	Seasonal differencing alone did not pass KPSS; residual seasonal structure remains.

Table 1. Stationarity test results.

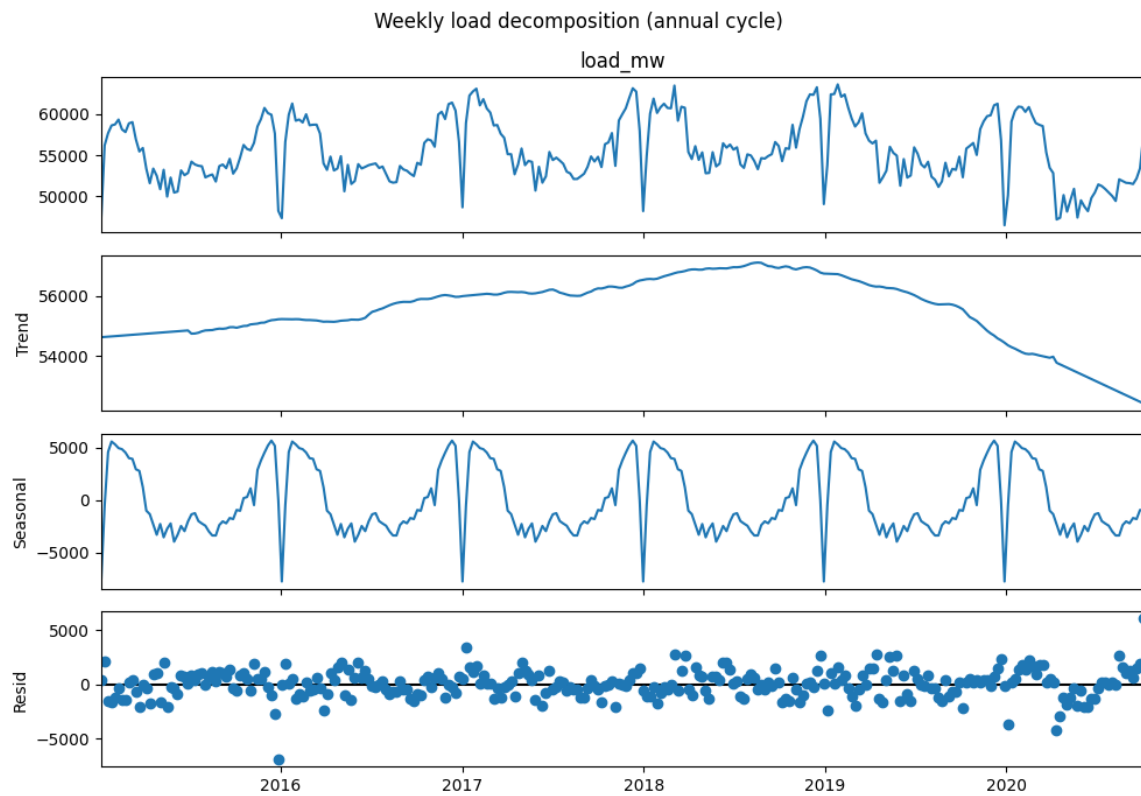


Figure 2. Weekly seasonal decomposition showing the annual component and residual variation.

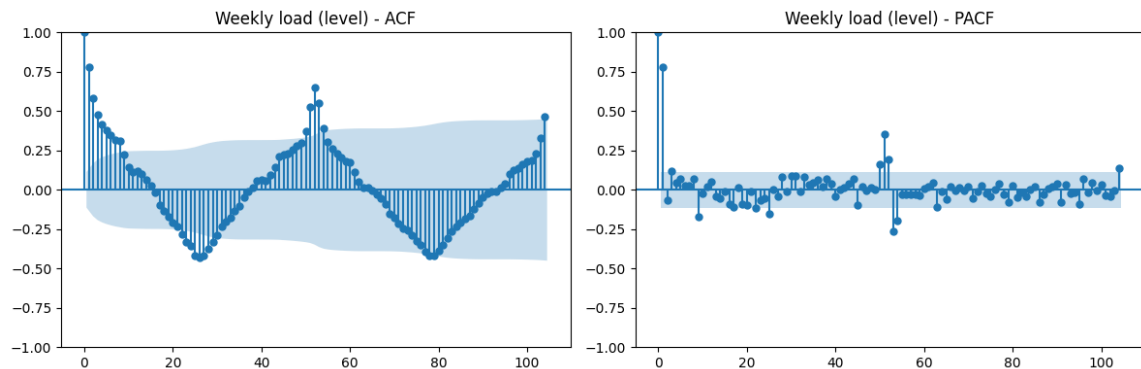


Figure 3. Weekly load ACF/PACF in levels, showing strong annual dependence.

ADF and KPSS together suggest that the weekly level series is not strongly non-stationary in the usual stochastic-trend sense. However, the ACF/PACF and decomposition indicate clear seasonal dependence. Therefore, the main modelling issue is not only differencing the mean level; it is representing the annual seasonal cycle and abnormal demand shocks.

2. Benchmark forecasts

Four benchmark methods were used: the historical mean, naive last-value forecast, seasonal naive forecast with a 52-week season, and drift forecast. These models provide a necessary baseline because complex models should only be preferred if they materially improve forecast accuracy or provide additional interpretability.

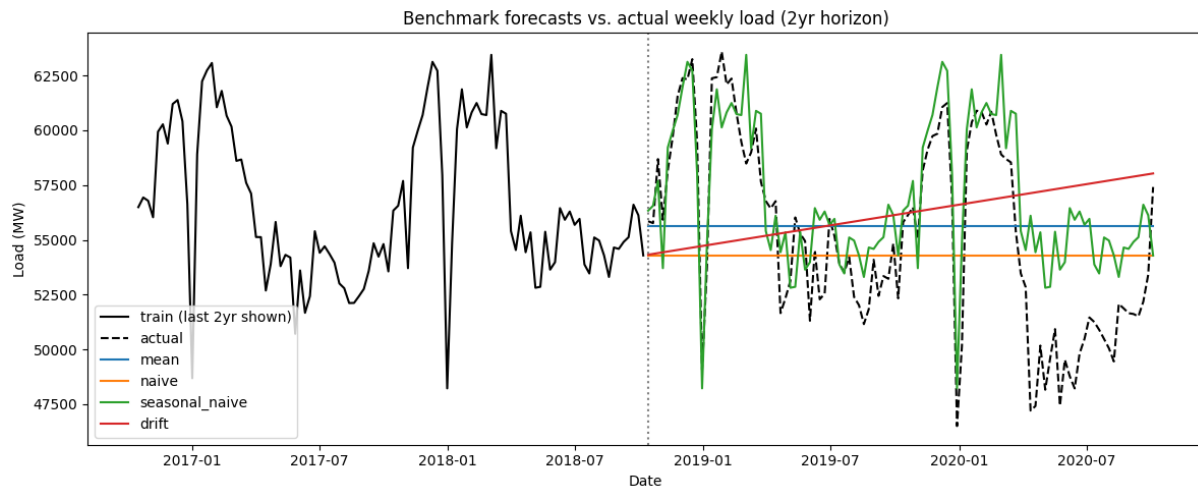


Figure 4. Benchmark forecasts against the final two years of weekly load.

Model	RMSE (MW)	MAE (MW)	MAPE (%)	Improvement vs seasonal naive
LSTM weekly aggregate	573.9	477.5	0.91	80.9%
Gradient Boosting	2853.4	2185.2	4.19	5.1%
Seasonal naive	3006.8	2318.5	4.41	0.0%
Random Forest	3021.1	2277.6	4.39	-0.5%
Mean	4397.3	3788.8	6.97	-46.2%
Naive	4459.1	3783.2	6.79	-48.3%
Drift	5118.0	4339.9	8.05	-70.2%
SARIMAX + temperature	7.81e17	3.49e17	6.12e14	Diverged
SARIMA	inf	2.20e160	3.91e157	Diverged

Table 2. Forecast evaluation on the final 104 weekly observations. LSTM hourly results were aggregated to weekly values for comparison.

3. SARIMA and SARIMAX modelling

Because the weekly series contains an annual seasonal component, the analysis used a SARIMA structure with seasonal period 52. AIC-based grid search over the fitted candidate set selected SARIMA(2,0,1)x(1,0,1,52), with AIC 1118.1. The chosen non-seasonal differencing order $d=0$ is consistent with the stationarity tests, where the weekly level series passed both ADF and KPSS at the 5% decision level. The seasonal period of 52 is justified by weekly aggregation and the annual demand cycle identified in the decomposition and ACF.

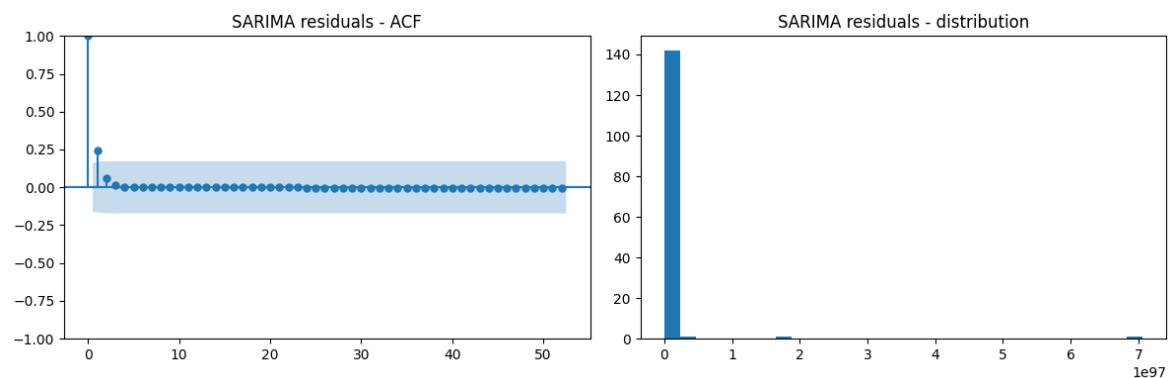


Figure 5. SARIMA residual ACF and residual distribution.

However, the selected SARIMA model was not acceptable after diagnostics. The model summary reported unstable coefficient behaviour, including a near-singular covariance matrix, a very large seasonal MA coefficient, non-normal residuals, and significant Ljung-Box autocorrelation. The residual mean and standard deviation were approximately $6.41e95$ and $6.03e96$ after the seasonal warm-up period. The forecast therefore exploded, giving infinite RMSE. This demonstrates a central limitation of AIC: it compares in-sample likelihoods but does not guarantee a stable, well-calibrated, or useful forecast.

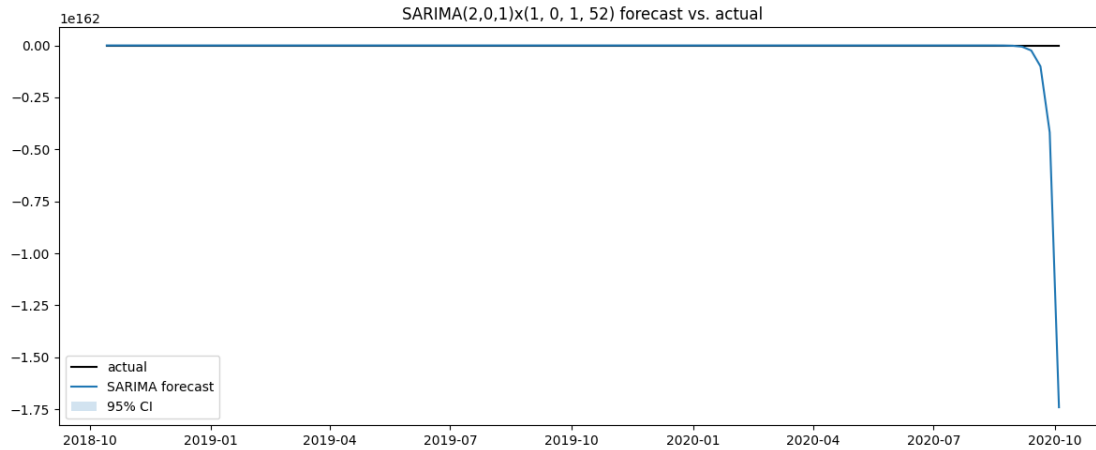


Figure 6. SARIMA forecast showing numerical divergence rather than a usable prediction.

The SARIMAX extension added Berlin weekly mean temperature as an exogenous regressor. Temperature is physically relevant because electricity demand is affected by heating and cooling behaviour. In the analysis, future observed temperature was supplied for the test period, so this is an explanatory or conditional forecast rather than a true operational forecast from the forecast origin. The SARIMAX model also diverged, with RMSE $7.81e17$ MW, so the temperature coefficient cannot be interpreted as reliable despite appearing statistically significant. A more robust SARIMAX specification would require scaling, alternative orders, coefficient constraints, residual checks and possibly holiday/calendar regressors.

4. Feature-based regression and leakage control

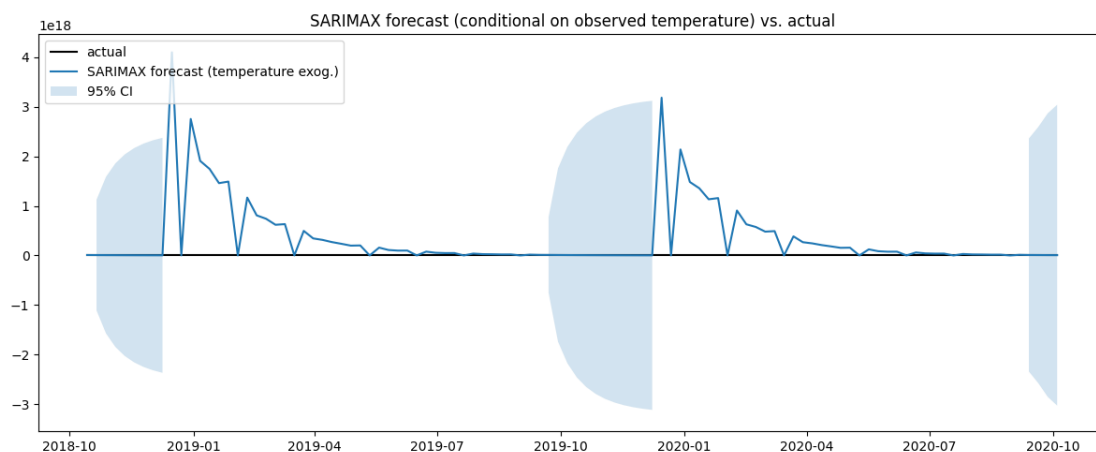


Figure 7. SARIMAX conditional temperature forecast; the instability means the model is rejected.

The regression models used weekly features: temperature, lag_1, lag_2, lag_52, a shifted four-week rolling mean, week-of-year sine/cosine terms and month. Random Forest and Gradient Boosting were trained on

the pre-test period and evaluated on the final two years. Leakage was controlled in the feature construction by shifting load-derived features before they were used as predictors: lag_1, lag_2, lag_52 and the rolling mean use only previous load values relative to the prediction row. This is correct for a rolling operational setting where recent observed load is available each week. For a pure two-year forecast made only at the first forecast origin, those future observed lags would not be available and recursive prediction would be required.

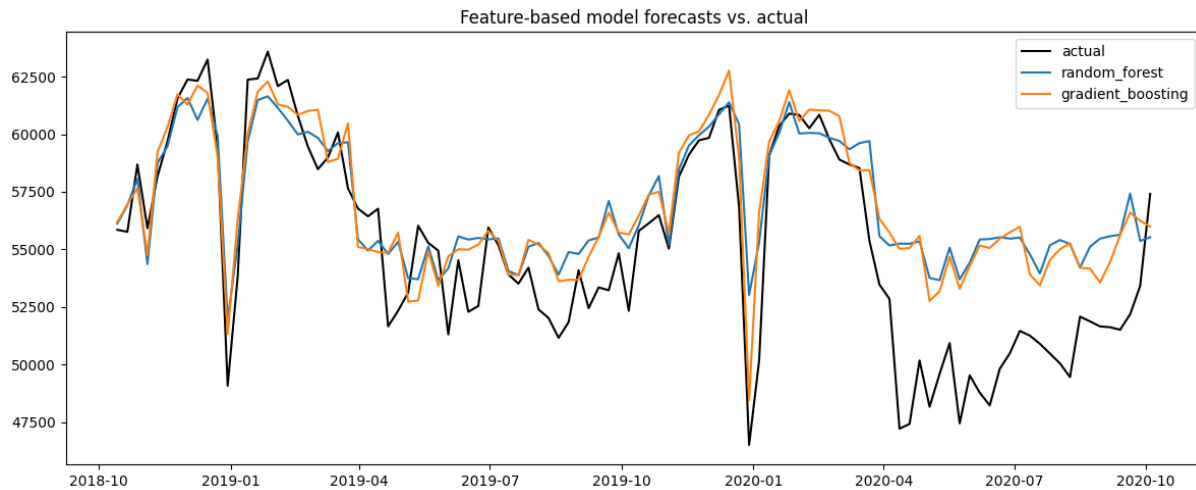


Figure 8. Random Forest and Gradient Boosting weekly forecasts.

The feature importances show that lag_52 dominates both tree-based models, contributing approximately 0.812 in Random Forest and 0.822 in Gradient Boosting. Temperature had modest importance, around 0.036-0.040. This explains the results: the models mainly reproduce the previous annual load shape and make only limited corrections using weather and short-term lags. Gradient Boosting slightly improved over seasonal naive because it can learn residual nonlinear adjustments, whereas Random Forest was marginally worse than seasonal naive, suggesting that additional flexibility did not translate into better extrapolation in the abnormal 2019-2020 test period.

5. LSTM modelling and literature context

The LSTM model was trained on hourly load using a 168-hour lookback window, corresponding to one week of hourly history. The architecture contained two LSTM layers with 64 and 32 units, dropout of 0.2, a dense hidden layer with 16 units and a final one-step output. The model had 29,857 trainable parameters. The model evaluation reported hourly test RMSE 1638.8 MW, MAE 1229.8 MW and MAPE 2.32%; after aggregating the hourly predictions to weekly means, the weekly RMSE reduced to 573.9 MW and MAPE to 0.91%.

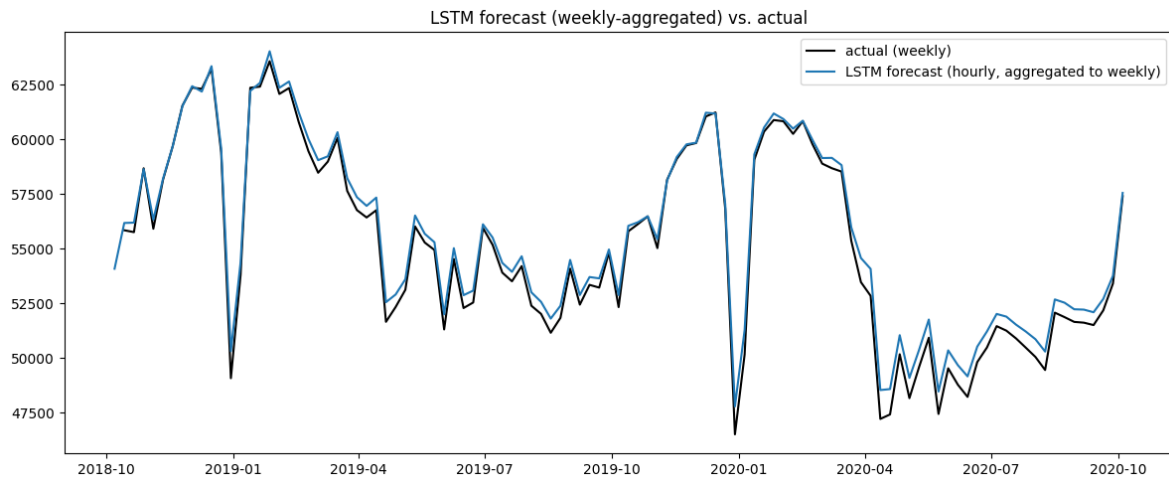


Figure 9. LSTM hourly forecast aggregated to weekly load.

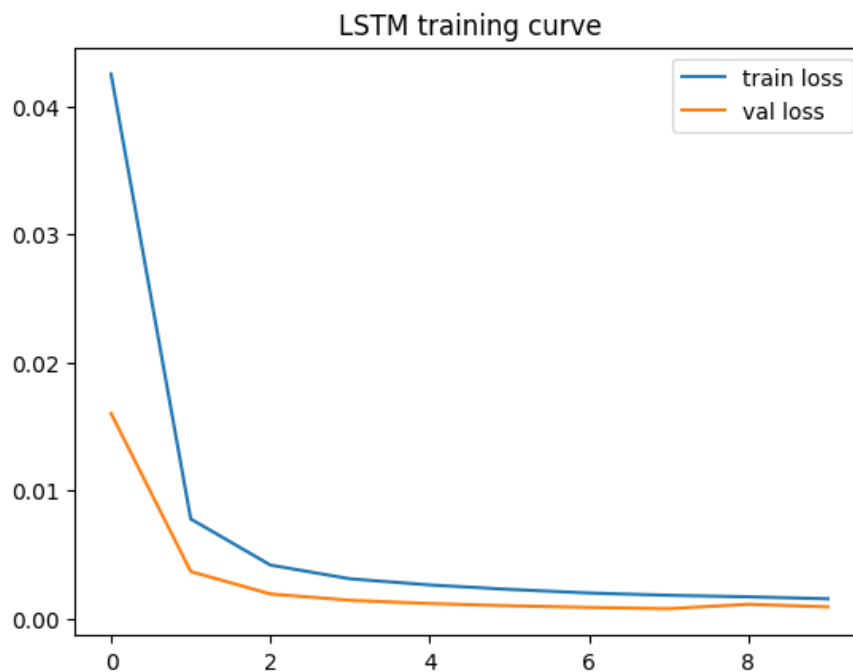


Figure 10. LSTM training and validation loss curve.

The strong LSTM result is consistent with the literature on recurrent neural networks for electricity load forecasting: LSTM networks are designed to handle temporal dependencies and mitigate the vanishing-gradient problem in sequence data. In load forecasting, this is useful because demand depends on recent load, weekly routines and seasonal patterns. However, the analysis used a one-step-ahead rolling-test design in which each test input window is constructed from true past observations. That is realistic for continuous monitoring but not the same as a fully open-loop two-year forecast issued once. Therefore, the LSTM achieves the best accuracy in this analysis, but it should be interpreted as rolling short-term forecasting performance.

6. Discussion: comparison, covariates and operational recommendation

Compared with the seasonal naive benchmark, only two models showed meaningful improvement: the weekly aggregate LSTM and Gradient Boosting. The LSTM reduced RMSE by 80.9%, while Gradient Boosting reduced it by 5.1%. Random Forest was effectively tied with seasonal naive and therefore does not justify its additional complexity. Mean, naive and drift forecasts performed poorly because they do not represent annual seasonality. SARIMA and SARIMAX were rejected because their forecasts diverged.

Temperature improves the physical realism of the modelling problem, but this analysis does not provide reliable evidence that it improved SARIMAX accuracy, as the fitted SARIMAX model was unstable. In the tree models, temperature had only modest feature importance, while the 52-week lag was much more influential. Holiday covariates were not implemented in the final model set, so no empirical claim can be made about their effect. Conceptually, holidays are useful because they are known in advance at the forecast origin and can explain demand reductions around public holidays; observed future temperature, in contrast, is not known unless supplied by a weather forecast.

For operational use, the recommended model is Gradient Boosting, not the LSTM, despite the LSTM having the lowest error. Gradient Boosting gives a small but real improvement over the seasonal naive benchmark, is easier to retrain and maintain, provides feature importance for interpretation, and avoids the numerical instability seen in SARIMA/SARIMAX. It is also more transparent than the LSTM. The LSTM would be recommended only if the operational requirement is rolling short-term forecasting with frequent retraining, proper validation, recursive multi-step testing and monitoring for drift. A production system should also retain the seasonal naive model as a mandatory benchmark and fallback.

7. Answers to assignment questions

1. Comparison against seasonal naive: The LSTM weekly aggregate and Gradient Boosting improved on seasonal naive. Random Forest did not provide a meaningful improvement, and the SARIMA/SARIMAX models were unusable due to divergence.

2. Avoiding leakage in temperature and lag features: Load lags and rolling means were shifted so the target week itself was not used as a predictor. Observed test-period temperature creates a conditional forecast; a true operational version must use the forecasted temperature available at the origin.

3. SARIMAX differencing and seasonal period: The level series passed ADF and KPSS, supporting $d=0$. Weekly aggregation gives a natural annual seasonal period of 52. Seasonal terms were needed because the decomposition and ACF showed strong annual dependence.

4. Temperature and holiday covariates: Temperature was included but did not produce a stable SARIMAX result; in tree models, it had modest importance. Holidays were not implemented and remain a limitation. Holidays are known in advance, whereas future temperatures are only known if a weather forecast is used.

5. Interpretability and complexity: SARIMAX is theoretically interpretable and provides intervals, but is sensitive to specification. Tree regressors are moderately interpretable through feature importance and easier to maintain. LSTM is the most accurate here but least interpretable and most maintenance-heavy.

6. Recommended operational model: Gradient Boosting is recommended for the current evidence base because it balances accuracy, interpretability and maintainability. LSTM should be considered for a rolling short-term system after stricter leakage-free multi-step validation.

8. Future improvements

Future work should run a constrained and diagnostically safe SARIMA/SARIMAX search rather than relying only on minimum AIC; add calendar and German public-holiday indicators; use weather forecasts rather than observed future temperature; test recursive multi-step forecasts for the machine-learning and LSTM models; perform rolling-origin cross-validation; and quantify forecast uncertainty for the regression and LSTM models, for example using quantile regression, bootstrapping or conformal prediction. These steps would make the analysis closer to a real operational forecasting system.

References

1. Box, G. E. P., Jenkins, G. M., Reinsel, G. C. and Ljung, G. M. (2015). Time Series Analysis: Forecasting and Control. 5th edn. Hoboken: Wiley.
2. Dong, Q., Zhang, Y., Liu, X. and Wang, H. (2024). Short-Term Electricity Load Forecasting by Deep Learning: A Comprehensive Survey. arXiv:2408.16202.
3. Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), pp. 1735-1780.
4. Hyndman, R. J. and Athanasopoulos, G. (2021). Forecasting: Principles and Practice. 3rd edn. Melbourne: OTexts.
5. Open Power System Data (2020). Time series data package, 2020-10-06 release. Available at: https://data.open-power-system-data.org/time_series/ (Accessed: 8 July 2026).
6. Open Power System Data (2020). time_series_60min_singleindex.csv, 2020-10-06 release. Available at: https://data.open-power-system-data.org/time_series/2020-10-06/time_series_60min_singleindex.csv (Accessed: 8 July 2026).
7. Open-Meteo (2026). Historical Weather API documentation. Available at: <https://archive-api.open-meteo.com/v1/archive> (Accessed: 8 July 2026).